Apache Pig: Bandwidth Utilization Analysis Using Pig Latin

Objective:

- Process raw web server access log data using Apache Pig.
- Extract the client IP address and response bytes from each log record.
- Group records by client IP address.
- Calculate the total bandwidth (response bytes) used by each IP address.
- Execute the Pig Latin program in **local mode** for a simple demonstration.

Scenario:

A web hosting provider stores raw web server access logs. The operations team wants to find how much data was transferred to each client IP address. Each record contains a client IP address and the number of response bytes transferred.

Problem Statement:

Write an Apache Pig program that:

- Reads the access log file.
- Extracts the client IP address and response bytes.
- Groups all records having the same IP address.
- Calculates the total response bytes for every IP address.

Mathematically: Total Bandwidth for an IP = SUM(response bytes for all requests from that IP).

Simple Demo Setup:

1. Install Java (JDK 8 or 11) and download Apache Pig 0.17.0.
2. Open a terminal in the Pig folder and use **local mode**; Hadoop is not required for this demo.
3. Create **access.log** and **bandwidth.pig** in the same folder.
4. Run: **pig -x local bandwidth.pig** and observe the output file.

Pig Latin Program:

```
logs = LOAD 'access.log' USING PigStorage(' ')
AS (ip:chararray, bytes:long);
grp = GROUP logs BY ip;
tot = FOREACH grp GENERATE group AS ip, SUM(logs.bytes) AS total_bytes;
STORE tot INTO 'output' USING PigStorage('\t');
```

Sample access.log: 192.168.1.10 500 192.168.1.20 300 192.168.1.10 700

Expected result: 192.168.1.10 → 1200 bytes 192.168.1.20 → 300 bytes

Software and Tools used:

Tool	Purpose
Java JDK 8/11	Runtime required by Apache Pig.
Apache Pig 0.17.0	Pig Latin data-processing framework.
Pig Local Mode	Runs the demo without setting up Hadoop.